

- 15.6 Using the NLS panel data on  $N = 716$  young women, we consider only years 1987 and 1988. We are interested in the relationship between  $\ln(WAGE)$  and experience, its square, and indicator variables for living in the south and union membership. Some estimation results are in Table 15.10.

TABLE 15.10 Estimation Results for Exercise 15.6

|           | (1)<br>OLS 1987     | (2)<br>OLS 1988     | (3)<br>FE           | (4)<br>FE Robust    | (5)<br>RE           |
|-----------|---------------------|---------------------|---------------------|---------------------|---------------------|
| $C$       | 0.9348<br>(0.2010)  | 0.8993<br>(0.2407)  | 1.5468<br>(0.2522)  | 1.5468<br>(0.2688)  | 1.1497<br>(0.1597)  |
| $EXPER$   | 0.1270<br>(0.0295)  | 0.1265<br>(0.0323)  | 0.0575<br>(0.0330)  | 0.0575<br>(0.0328)  | 0.0986<br>(0.0220)  |
| $EXPER^2$ | -0.0033<br>(0.0011) | -0.0031<br>(0.0011) | -0.0012<br>(0.0011) | -0.0012<br>(0.0011) | -0.0023<br>(0.0007) |
| $SOUTH$   | -0.2128<br>(0.0338) | -0.2384<br>(0.0344) | -0.3261<br>(0.1258) | -0.3261<br>(0.2495) | -0.2326<br>(0.0317) |
| $UNION$   | 0.1445<br>(0.0382)  | 0.1102<br>(0.0387)  | 0.0822<br>(0.0312)  | 0.0822<br>(0.0367)  | 0.1027<br>(0.0245)  |
| $N$       | 716                 | 716                 | 1432                | 1432                | 1432                |

(standard errors in parentheses)

- The OLS estimates of the  $\ln(WAGE)$  model for each of the years 1987 and 1988 are reported in columns (1) and (2). How do the results compare? For these individual year estimations, what are you assuming about the regression parameter values across individuals (heterogeneity)?
- The  $\ln(WAGE)$  equation specified as a panel data regression model is

$$\ln(WAGE_{it}) = \beta_1 + \beta_2 EXPER_{it} + \beta_3 EXPER_{it}^2 + \beta_4 SOUTH_{it} + \beta_5 UNION_{it} + (u_i + e_{it}) \quad (XR15.6)$$

- Explain any differences in assumptions between this model and the models in part (a).
- Column (3) contains the estimated fixed effects model specified in part (b). Compare these estimates with the OLS estimates. Which coefficients, apart from the intercepts, show the most difference?
- The  $F$ -statistic for the null hypothesis that there are no individual differences, equation (15.20), is 11.68. What are the degrees of freedom of the  $F$ -distribution if the null hypothesis (15.19) is true? What is the 1% level of significance critical value for the test? What do you conclude about the null hypothesis.
- Column (4) contains the fixed effects estimates with cluster-robust standard errors. In the context of this sample, explain the different assumptions you are making when you estimate with and without cluster-robust standard errors. Compare the standard errors with those in column (3). Which ones are substantially different? Are the robust ones larger or smaller?
- Column (5) contains the random effects estimates. Which coefficients, apart from the intercepts, show the most difference from the fixed effects estimates? Use the Hausman test statistic (15.36) to test whether there are significant differences between the random effects estimates and the fixed effects estimates in column (3) (Why that one?). Based on the test results, is random effects estimation in this model appropriate?

a.

The OLS estimates for 1987 and 1988 are very similar, without any significant difference.

By running separate OLS regressions for each year, we are assuming that any unobserved individual heterogeneity is part of the random error term, and this unobserved heterogeneity is uncorrelated with the explanatory variables in the model.

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b.

The error term is decomposed into  $u_i + e_{i,t}$ . The model distinguishes unobserved, time-invariant individual heterogeneity ( $u_i$ ) from the idiosyncratic error ( $e_{i,t}$ ). This indicates a change in our assumption: instead of assuming all unobserved factors are just random noise, this panel framework isolates these individual differences.

c.

Comparing the estimates of column 3 and column 1 & 2, I find that the coefficients of EXPER and SOUTH change dramatically. One of the characteristics of FE is that it eliminates  $u_i$  by the Difference or Within Estimator. Because the unobserved heterogeneity ( $u_i$ ) is correlated with these explanatory variables, OLS suffers from omitted variable bias. FE removes this bias, resulting in the significant difference in the estimated coefficients.

d.

$H_0$ : There are no individual differences.

$$df_1 = N - 1 = 916 - 1 = 915$$

$$df_2 = NT - N - K_0 \text{ (随时间变动的解释变量个数)} = 916 \times 2 - 916 - 4 = 912$$

$$F(0.01, 915, 912) \approx 1$$

The test statistic  $F = 11.68$  is much greater than the critical value, it falls in the rejection region. Therefore, we reject the null hypothesis. This implies that there are significant unobserved individual differences.

e.

When using the standard FE model (column 3), we assume the idiosyncratic error ( $\epsilon_{i,t}$ ) is homoskedastic and has no serial correlation. However, estimating with cluster-robust standard errors (column 4) relaxes this assumption, allowing for potential heteroskedasticity and within-cluster serial correlation.

Comparing the two columns, the standard error of SOUTH is substantially different. The robust standard errors are generally larger, indicating that ignoring cluster correlation would lead to underestimated standard errors.

f.

$$\text{Hausman test statistic} \Rightarrow t = \frac{b_{FE, \text{SOUTH}} - b_{RE, \text{SOUTH}}}{[\hat{\text{var}}(b_{FE, k}) - \hat{\text{var}}(b_{RE, k})]^{\frac{1}{2}}} = \frac{b_{FE, k} - b_{RE, k}}{[\text{se}(b_{FE, k})^2 - \text{se}(b_{RE, k})^2]^{\frac{1}{2}}}$$

$$t = \frac{-0.3261 - (-0.2326)}{\sqrt{0.1238^2 - 0.0317^2}} = -0.768$$

Since  $|-0.768| < |1.96|$ , there isn't a significant difference between the estimates. We can simply use random effect model.

**15.20** This exercise uses data from the STAR experiment introduced to illustrate fixed and random effects for grouped data. In the STAR experiment, children were randomly assigned within schools into three types of classes: small classes with 13–17 students, regular-sized classes with 22–25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file *star*.

- a. Estimate a regression equation (with no fixed or random effects) where *READSCORE* is related to *SMALL*, *AIDE*, *TCHEXPER*, *BOY*, *WHITE\_ASIAN*, and *FREELUNCH*. Discuss the results. Do students perform better in reading when they are in small classes? Does a teacher's aide improve scores? Do the students of more experienced teachers score higher on reading tests? Does the student's sex or race make a difference?
- b. Reestimate the model in part (a) with school fixed effects. Compare the results with those in part (a). Have any of your conclusions changed? [Hint: specify *SCHID* as the cross-section identifier and *ID* as the "time" identifier.]
- c. Test for the significance of the school fixed effects. Under what conditions would we expect the inclusion of significant fixed effects to have little influence on the coefficient estimates of the remaining variables?
- d. Reestimate the model in part (a) with school random effects. Compare the results with those from parts (a) and (b). Are there any variables in the equation that might be correlated with the school effects? Use the LM test for the presence of random effects.
- e. Using the *t*-test statistic in equation (15.36) and a 5% significance level, test whether there are any significant differences between the fixed effects and random effects estimates of the coefficients on *SMALL*, *AIDE*, *TCHEXPER*, *WHITE\_ASIAN*, and *FREELUNCH*. What are the implications of the test outcomes? What happens if we apply the test to the fixed and random effects estimates of the coefficient on *BOY*?
- f. Create school-averages of the variables and carry out the Mundlak test for correlation between them and the unobserved heterogeneity.

Coefficients:

|                | Estimate  | Std. Error | t value  | Pr(> t )           |
|----------------|-----------|------------|----------|--------------------|
| (Intercept)    | 437.76425 | 1.34622    | 325.180  | < 2e-16 ***        |
| small          | 5.82282   | 0.98933    | 5.886    | 4.19e-09 ***       |
| aide           | 0.81784   | 0.95299    | 0.858    | 0.391              |
| tchexper       | 0.49247   | 0.06956    | 7.080    | 1.61e-12 ***       |
| boy            | -6.15642  | 0.79613    | -7.733   | 1.23e-14 ***       |
| white_asian    | 3.90581   | 0.95361    | 4.096    | 4.26e-05 ***       |
| freelunch      | -14.77134 | 0.89025    | -16.592  | < 2e-16 ***        |
| ---            |           |            |          |                    |
| Signif. codes: | 0 '***'   | 0.001 '**' | 0.01 '*' | 0.05 '.' 0.1 ' ' 1 |

a.

- Small classes:
  - The coefficient of SMALL is **positive and significant**. This indicates that, on average, students perform better in reading when they are assigned to small classes.
- Teacher' s aide:
  - The coefficient of AIDE is **positive but statistically insignificant**. This suggests that having a teacher' s aide does not have a significant impact on students' reading scores.
- Teacher experience:
  - The coefficient of TCHEXPER is **positive and significant**. This indicates that students taught by more experienced teachers score higher on their reading tests.
- Student' s sex:
  - The coefficient of BOY is **negative and significant**. This means that male students, on average, score lower on reading tests.
- Student' s race:
  - The coefficient of WHITE\_ASIAN is **positive and significant**, indicating that White and Asian students, on average, score higher on reading tests.

b.

- Small classes:
  - The coefficient of SMALL **increased** in the fixed effect model, and it remain significant. **Conclusion has not changed.**
- Teacher' s aide:
  - The coefficient of AIDE remains statistically insignificant. **Conclusion has not changed:** A teacher' s aide does not significantly improve reading scores.
- Teacher experience:
  - The coefficient of TCHEXPER **decreased** in the fixed effect model, though it remains significant. This suggests that part of the "experience premium" in part (a) was due to experienced teachers sorting into better-performing schools. **Once we control for the school effect, the pure impact of teacher experience is smaller but still positive.**
- Student' s sex:
  - The coefficient of BOY remains **negative** in the fixed effect model, and it is still significant. **Conclusion has not changed.**
- Student' s race:
  - There is a **dramatic change** in the coefficient of WHITE\_ASIAN. This substantial change implies that a student' s race is **highly correlated with the unobserved school characteristics**. By controlling for school fixed effects, we isolate the within-school racial achievement gap, which is significantly larger than what the pooled OLS model suggested.

#### F test for individual effects

c.

```
data: readscore ~ small + aide + tchexper + boy + white_asian +
freelunch
F = 16.698, df1 = 78, df2 = 5681, p-value < 2.2e-16
alternative hypothesis: significant effects
```

- $H_0: \beta_{11} = \beta_{12} = \dots = \beta_{1N}$  (Pooled OLS is enough.)
- $H_1$ : the  $\beta_{1i}$  are not all equal.
- We perform an F-test to test the joint significance of the school fixed effects. The test yields an **F-statistic of 16.698** with a **p-value < 2.2e-16**.
- Since the p-value is essentially zero and much less than the standard 0.05 significance level, we **reject the null hypothesis**. This confirms that the school fixed effects are highly significant, and the Fixed Effects model in part (b) is statistically superior to the Pooled OLS model in part (a).
- We would expect the inclusion of these significant fixed effects to have **little influence** on the coefficient estimates of the remaining variables **only if the unobserved school characteristics are completely uncorrelated with the explanatory variables** (such as class size, race, or teacher experience). However, as we saw in part (b) with the WHITE\_ASIAN variable, there is a clear correlation, which is why controlling for fixed effects is necessary to avoid omitted variable bias.

d.

- Estimating the model with random effects yields coefficients that fall between the Pooled OLS and Fixed Effects estimates.
- The coefficient of WHITE\_ASIAN in the RE model is 7.35, which is a substantial jump from the Pooled OLS estimate and closer to the FE estimate. Since the FE model strictly controls for school-level unobserved heterogeneity while the RE model assumes no correlation, **this dramatic change implies that race is highly correlated with the unobserved school characteristics**.
- Besides, we conduct the LM test to check for the presence of random effects. The null hypothesis is that the variance of the unobserved school effects is zero. The test yields an **extremely small p-value**, concluding that **significant random school effects are present and a panel data is appropriate**.

#### Lagrange Multiplier Test - (Honda)

```
data: readscore ~ small + aide + tchexper + boy + white_asian +
freelunch
normal = 81.715, p-value < 2.2e-16
alternative hypothesis: significant effects
```

e.

- Looking at the calculated statistics, **the absolute values for all five variables are less than 1.96** (the largest being tchexper). Therefore, none of the individual coefficients show a statistically significant difference between the FE and RE models at the 5% level.
- If we attempt to apply the same test to the **boy coefficient**, the calculation fails mathematically and returns **NaN** (Not a Number).
- This occurs because, in this specific sample, the estimated standard error for the Random Effects model is slightly larger than the standard error for the Fixed Effects model. Consequently, the term inside the square root in the denominator evaluates to a **negative number**.
- From an econometric perspective, this is a classic **finite-sample anomaly**. The standard Hausman test relies on asymptotic theory, which assumes that under the null hypothesis, the RE estimator is fully efficient and thus. However, in finite samples, this theoretical variance inequality does not

```
> print(t_stats)
      small      aide    tchexper  white_asian  freelunch
1.14600764  0.12843803 -1.93771666  1.21807432 -0.09555102
```

always hold strictly. When this anomaly occurs, it technically violates the assumptions required to compute this specific form of the test statistic, rendering the classical test inconclusive for this individual variable.

f.

- A joint linear hypothesis test (Wald test) was conducted to determine if the coefficients on these school-average variables were jointly zero. **The test yields a Chi-square statistic of 13.38 with a p-value of 0.03738. Since the p-value is less than 0.05, we reject the null hypothesis.**
- This outcome confirms that **the unobserved school-level heterogeneity is indeed correlated with the explanatory variables in our model.** Therefore, the core assumption of the Random Effect model is violated. We can conclude that **Fixed Effects estimator is consistent and is preferred and appropriate model** for this experiment.

```
Model 1: restricted model
Model 2: readscore ~ small + aide + tchexper + boy + white_asian
+ freelunch +
    mean_small + mean_aide + mean_tchexper + mean_boy + mean_whit
e_asian +
    mean_freelunch

    Res.Df Df Chisq Pr(>Chisq)
1      5759
2      5753  6 13.38    0.03738 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```